

Group Project

STAT 300 (Fall 2024)

UNAUTHORIZED DISTRIBUTION AND/OR UPLOADING OF THIS DOCUMENT IS STRICTLY PROHIBITED.

Logan wants to open a baby store chain with the hope of expanding his business overseas. He is interested in finding out whether the average sales are associated with the characteristics of the store location. To understand this relationship, he gathered data on car seat sales for 500 locations. Attributes include:

Sales-Sales (in thousands of dollars) at each location

Price-Price company charges for car seats at each site

CompPrice - Price charged by competitor at each location

Income - Median income in the community (in 1000's of dollars)

Education - Education level at the location

Age - Average age of the local population

Advertising - Local advertising budget for company at each location (in thousands of dollars)

Accidents - Average monthly accidents in the area

Urban- A factor with levels: 0 and 1 to indicate whether the store is in an urban or rural location (1 if urban 0 if rural)

US - A factor with levels 0 and 1 to indicate whether the store is in the US or not (1 if in US 0 if other)

ShelveLoc - A factor with levels Bad, Good and Medium indicating the quality of the shelving location for the car seats at each site

Logan is seeking your assistance to build a model to make inferences and predictions about the car seat sales. To achieve this goal, you are expected to do below:

(1). First you are going to split the dataset into two: training and testing set. Use 400 observations in the dataset to train your models and remaining 100 to validate the prediction performance. When subsampling 400 observations for the training data, I want you to use a random seed. This random seed MUST have the format "300GroupNo". For instance, if your group is 1, the your random seed should be 30001. If your group number is 11 then, your random seed should be 30011, etc. Remember, for descriptive analysis as well as for developing the model, you are going to use your training set only. You will use the testing set to validate the prediction performance of the developed model.

You may adapt a code like below to split the data into testing and training set:

```
#Read the data
Cseat<-read.csv("carseat.csv")

#Set the seed following the format I have given above. For instance if you are in group 1 your seed should be 1234

set.seed(1234)

#Decide which records to keep in the training set. I sort them for the convenience
trainRec<-sort(sample(1:nrow(Cseat),size=400))
#Extract the indexes that were not in trainRec to the test records
testRec<-setdiff(1:nrow(Cseat),trainRec)

#Extract indexed rows in trainRec and keep them in the training set
trainSeat<-Cseat[trainRec,]
#Extract indexed rows in testRec and keep them in the training set
testSeat<-Cseat[testRec,]
```

- (2) You should pick a model using variable selection approaches such as stepwise, best subset with different information criteria such as AIC, BIC, etc. To pick the model, you are going to use your training set. You must clearly justify the reasons for your suggested model, which include model assumptions as well.
- (3) Based on all your observations, you have to recommend one model to your client. Clearly identify the important variables when it comes to determining the sales based on your developed model.
- (4) Finally, you must also illustrate the prediction ability of the developed model using testing set. Think of creative ways to convey this to your client.

As a group, you are expected to perform a statistical analyses to achieve the aforementioned objectives and prepare a summary report. Your report should contain your data analysis steps to cover above; how you arrived at your final model should be clearly explained. In doing so, you need to select which graphs and tables are essential to be included in the report. All figures and tables must be labeled and hyperlinks should be utilized when referring to the relevant tables/graphs. When you include a table, it must be a formatted table, not an R output. You should include the corresponding output in the appendix of the report. Be creative in your final remarks.

The main text of your report should be formatted to a **maximum of 4 pages** (standard letter size), 1.5 line spacing with 1 inch margin. Font type must be 10pt-Arial.

Select most important figures and tables to be presented. Any additional methods, tables and figures can be submitted as an appendix. Appendix does not have a page limit. Make sure to acknowledge any published resources you used, and your reference list can be outside the 4-page limit as well. Your reports should be submitted as a single PDF file. **Please note that only the first 4 pages of your report will be graded and appendix will be only be referred in special circumstances. All main results should be presented within the 4-page limit.**

Your short report should include:

- within the 4-page limit:
 - Background and Introduction (what is the research question you are trying to answer and why is it interesting, relevant information about the data, etc.)
 - Exploratory data analysis: Include any graphical and numerical summaries. Just including graphs/tables are not sufficient. You must explain what insight they give you into the statistical analysis.
 - Statistical Methods (how you arrived your final model from the variable selection approaches, your observations regarding assumptions, validation of the predictions)

using test set, etc.)

- Conclusion (Your final recommendations summarized based on the statistical analysis)
- Discussion (limitations, future directions, etc.)
- outside the page limit:
 - References
 - Appendix (if needed): any additional figures and tables, methods, etc.

Everyone in the group **MUST** put equal amount of time and effort to the project and it is your responsibility to let me know if someone in your group is not participating. Each group **MUST** submit a document signed by all the members in your group with how each member contributed to the project. **Please note that this is a group project, while each one of you may have their own contribution, everyone in the group is equally responsible for the final outcome of the project. For instance, if there is a mistake, you cannot simply put the blame on one person who worked on that particular component. Everyone is equally responsible for all content in your final product.**

Project is due by 11:55 pm (ET) on Wednesday Nov 13th. One member from the group must submit three documents in Canvas: your group report (as a single PDF), R/Rmd script file, and the e-signed contribution disclosure. The disclosure needs to e-signed by all members of the group by typing their full name. Please do not email me separate R codes from each member in the group. Make sure your R script file is properly consolidated and organized, and include comments so that I can understand without having to contact group members.